

FINANCIAL CONSUMER COMPLAINT ANALYSIS

1. Project Overview and Objective

The objective of this project is to analyze financial consumer complaint data and develop an interactive Power BI dashboard that provides actionable insights into customer grievance patterns, complaint volumes, product-related issues, response efficiency, and resolution status.

2. Data Sources

- **Source Description and Timeline:** Maven Analytics – Financial Customer Complaints **and** Time Line: 2017 to 2023
- **Domain:** Financial Services Domain.
- **Data Size:** 62,516 Rows and 14 Columns

3. Objectives

- To analyze financial consumer complaint data and identify major complaint trends across products, services, and issue categories.
- To monitor key performance indicators (KPIs) such as total complaints, timely responses, dispute rates, and resolution status for measuring operational efficiency.
- To evaluate customer grievance patterns across states, submission channels, and time periods to support data-driven business decisions.
- To identify high-risk products and recurring customer issues in order to improve service quality and customer satisfaction.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Complaint ID	Integer / String	Unique identifier assigned to each consumer complaint record
Product	Categorical (Text)	Financial product category associated with the complaint, such as credit card, mortgage, or loan services
Issue	Categorical (Text)	Primary issue or problem reported by the consumer regarding the financial product or service
Timely Response?	Categorical (Yes/No)	Indicates whether the company responded to the complaint within the expected service-level timeline

5. Tools & Technologies

- **Excel:** Used as the primary data source for importing and managing financial consumer complaint records
 - **Power BI:** Used for interactive dashboard development, data visualization, KPI reporting, and business intelligence analysis.
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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

1. DATA CLEANING IN EXCEL

- Initial dataset inspection
- Basic validation
- Understanding raw data structure

A. Dataset Inspection

- Reviewed all columns
- Understood attribute meanings
- Identified missing values
- Checked categorical fields
- Observed inconsistent records

B. Handling Missing Values

- What we checked:
- Blank rows
- Null values
- Empty categorical fields

C. Duplicate Record Identification

Checked for:

- Repeated Complaint IDs
- Duplicate complaint entries

D. Data Consistency Validation

- Standardized inconsistent text values.

E. Date Format Verification

- Ensured date columns were valid.

2. DATA CLEANING IN POWER QUERY

A. Removing Unnecessary Columns

Removed Columns not required for analysis.

B. Changing Data Types

Column	Data Type
Complaint ID	Whole Number
Date Received	Date
Product	Text
Timely Response?	Text
Resolution Days	Numeric

C. Standardizing Categorical Values

I Cleaned

- Product names
- Issue categories
- Response values

D. Null Value Handling

- Replaced nulls
- Removed incomplete records where necessary

E. Creating Derived Date Columns

Created:

- Month
- Year
- Quarter

F. Removing Errors

I Handled

- Invalid dates
- Formatting issues
- Conversion errors

G. Data Transformation

Applied:

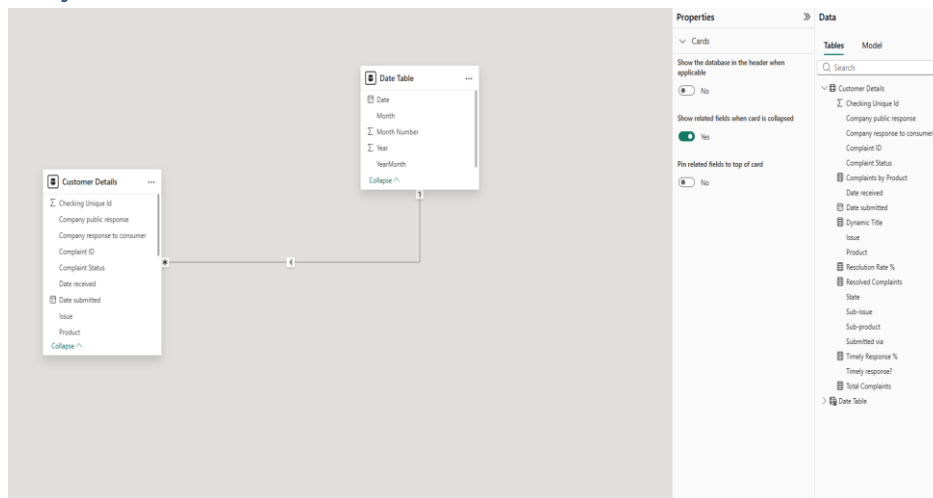
- Column renaming
- Formatting
- Trimming spaces
- Text cleanup

H. Creating Calculated Columns:

Created additional fields required for KPI reporting and analytical insights.

7. Data Modelling and DAX (Power BI)

- **Data Model:** Built relationships between tables to create a structured and scalable analytical data model

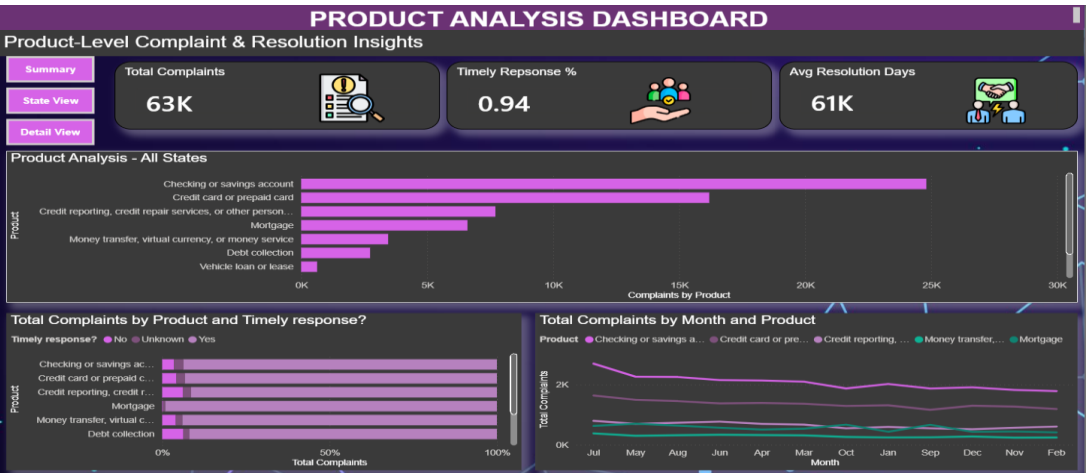
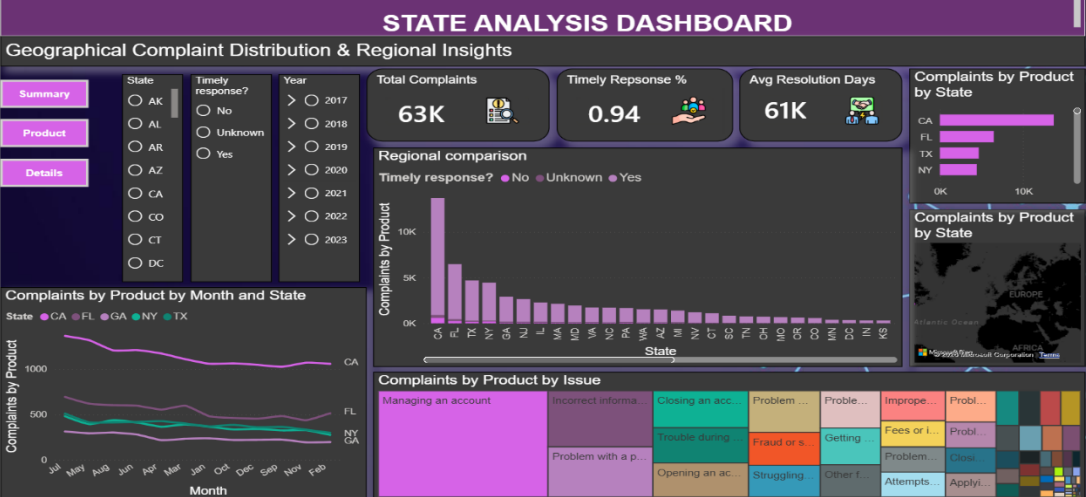
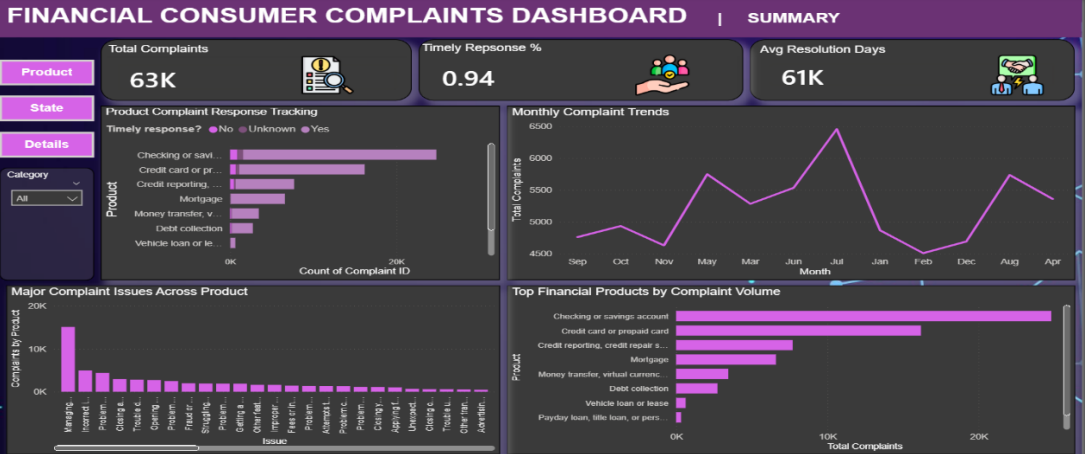


- **Calculated Columns & DAX Measures:** Developed dynamic DAX measures for KPI calculations and interactive reporting. Total complaints, Timely Response%, Resolved Complaints, Resolution Rate %, Dynamic Tiles, Complaints by Products.

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **Multiple Visualizations based on problem statement:** **Line Chart:** Used for time-series trend analysis to monitor complaint patterns across months and years, **Clustered Bar Chart:** Used to compare complaint volumes across different financial products and states, **Map Visualization:** Used for Geographical Analysis of complaint distribution across states
- **Slicers:** Used for interactive filtering based on Product, State, Date, and Response Status
- **Drillthrough Functionality:** Used to navigate from summarized insights into detailed complaint-level records
- **Navigation Buttons:** Used to create multi-page dashboard navigation and improve user experience

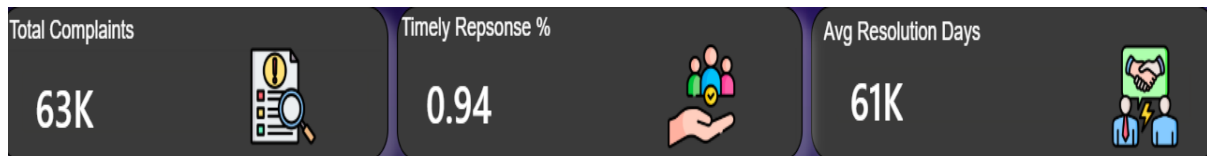


Complaint-Level Detail Analysis

Product View	Product	Complaint ID	Product	Issue	State	Timely Response %
State View	Checking or savings account					
Product	Closing an account					
Checking or savings account	2474932	2474932	Checking or savings account	Closing an account	FL	1.00
Total	2474932		Checking or savings account	Closing an account	FL	1.00
		2475326	Checking or savings account	Closing an account	FL	1.00
		2475485	Checking or savings account	Closing an account	FL	1.00
		2476071	Checking or savings account	Closing an account	AZ	1.00
		2476673	Checking or savings account	Closing an account	RI	1.00
		2477243	Checking or savings account	Closing an account	FL	1.00
		2477264	Checking or savings account	Closing an account	TX	1.00
		2478866	Checking or savings account	Closing an account	TX	1.00
		2479204	Checking or savings account	Closing an account	FL	1.00
		2480739	Checking or savings account	Closing an account	FL	1.00
		2482405	Checking or savings account	Closing an account	TX	1.00
		2483159	Checking or savings account	Closing an account	AZ	1.00
		2484208	Checking or savings account	Closing an account	FL	1.00
		2484524	Checking or savings account	Closing an account	FL	1.00
		2484923	Checking or savings account	Closing an account	DE	1.00
		2486130	Checking or savings account	Closing an account	WI	1.00
		2487422	Checking or savings account	Closing an account	CA	1.00
		2487592	Checking or savings account	Closing an account	MA	1.00
		2488474	Checking or savings account	Closing an account	CA	1.00
		2490682	Checking or savings account	Closing an account	MI	1.00
		2490718	Checking or savings account	Closing an account	OH	1.00
		2490723	Checking or savings account	Closing an account	NY	1.00
		2491636	Checking or savings account	Closing an account	MI	1.00
		2491673	Checking or savings account	Closing an account	GA	1.00
		2491740	Checking or savings account	Closing an account	AZ	1.00
		2492010	Checking or savings account	Closing an account	CA	1.00
		Total				0.94

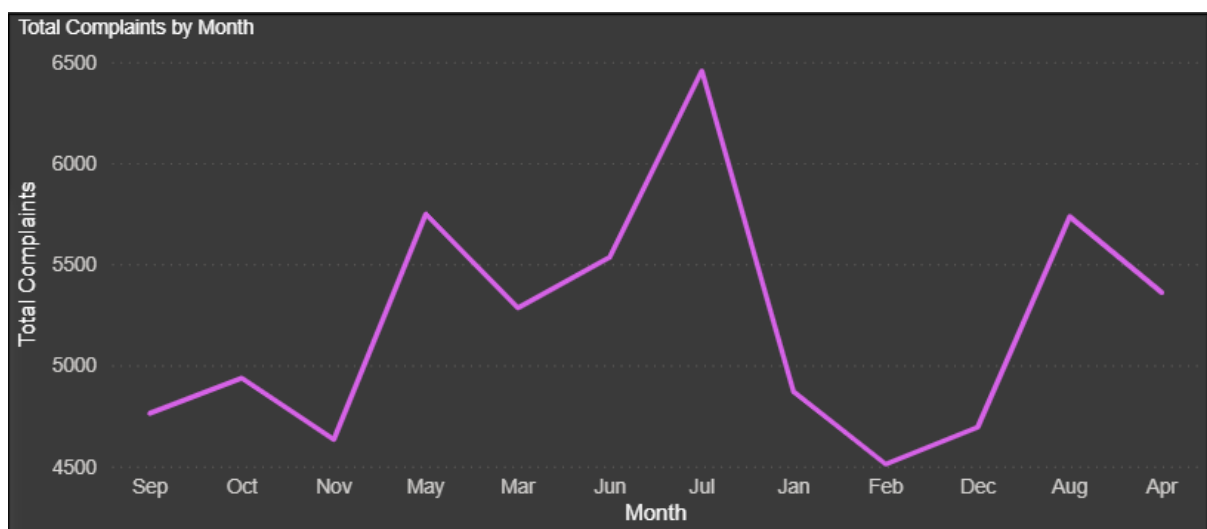
KPI Cards

KPI Cards were used to provide executive-level monitoring of key operational metrics such as Total Complaints, Timely Response Percentage, and Average Resolution Days, enabling quick performance evaluation and high-level business insight generation.



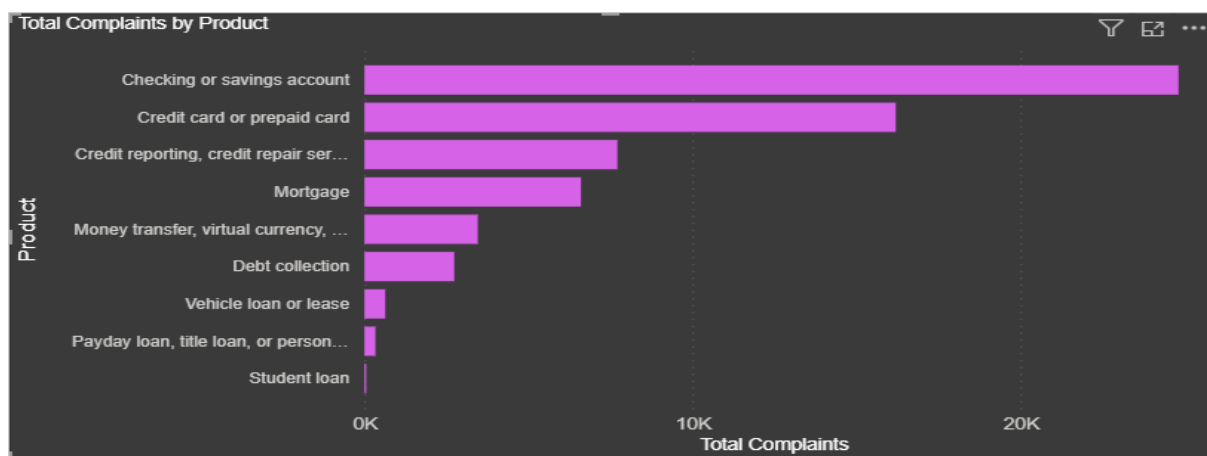
Line Chart (Complaint Trend Analysis)

The Line Chart was implemented to perform time-series analysis and monitor complaint fluctuations across different reporting periods, helping identify recurring complaint spikes, seasonal variations, and operational trend patterns.



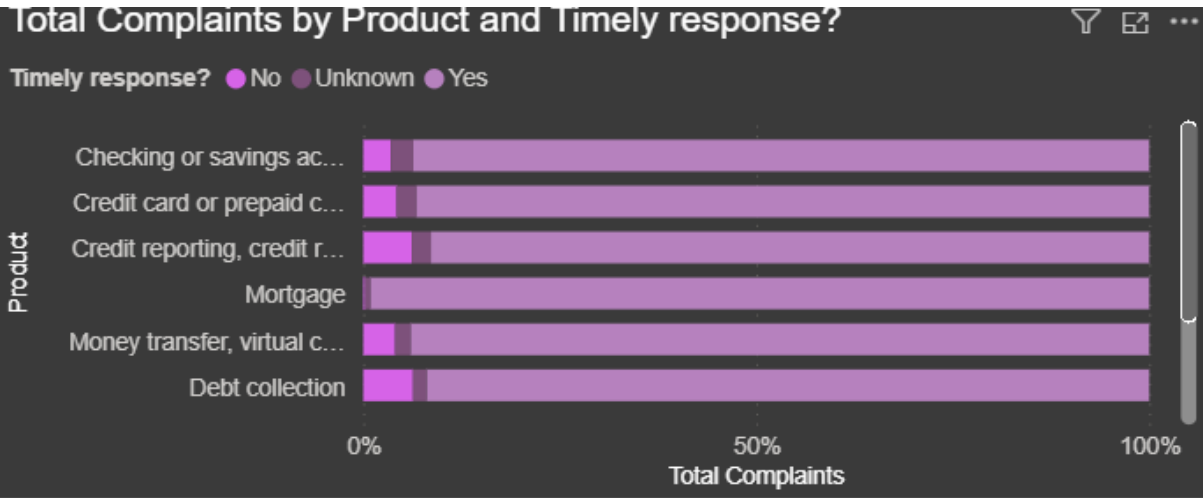
Clustered Bar Chart (Complaints by Product)

The Clustered Bar Chart was used to compare complaint volume across financial product categories and identify high complaint-generating products contributing to operational risk concentration.



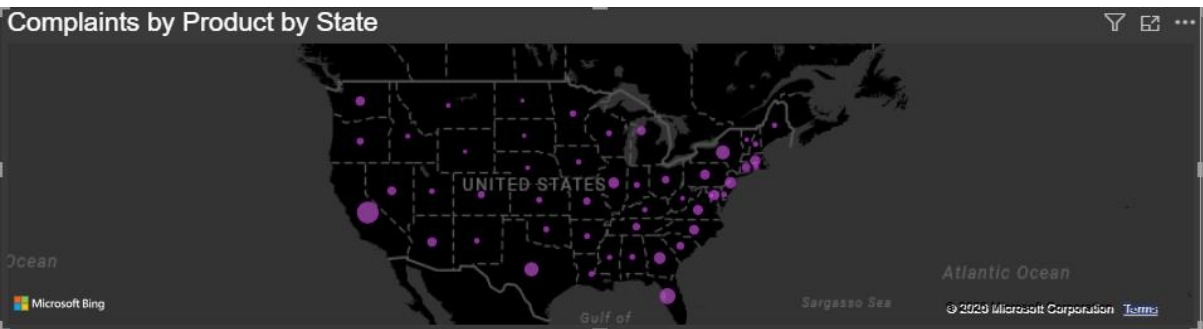
Stacked Bar Chart (Timely Response Analysis)

The Stacked Bar Chart was utilized to evaluate response performance across product categories by comparing timely and delayed responses, enabling operational efficiency assessment and SLA monitoring.



Map Visualization (Geographical Complaint Distribution)

The Map Visualization was implemented to perform geographical intelligence analysis and identify state-wise complaint concentration hotspots through spatial complaint distribution monitoring.



Treemap / Issue Analysis Visualization

The Treemap visualization was used to analyze complaint issue distribution and identify recurring customer pain points, enabling issue-category concentration analysis and operational problem identification.



Slicers

Slicers were integrated to support dynamic filtering and interactive dashboard exploration across dimensions such as Product, State, Date, and Response Status.

State	Timely response?	Year
☐ AK	☐ No	☐ 2017
☐ AL	☐ Unkno...	☐ 2018
☐ AR	☐ Yes	☐ 2019
☐ AZ		☐ 2020
☐ CA		☐ 2021
☐ CO		☐ 2022
☐ CT		☐ 2023
☐ DC		

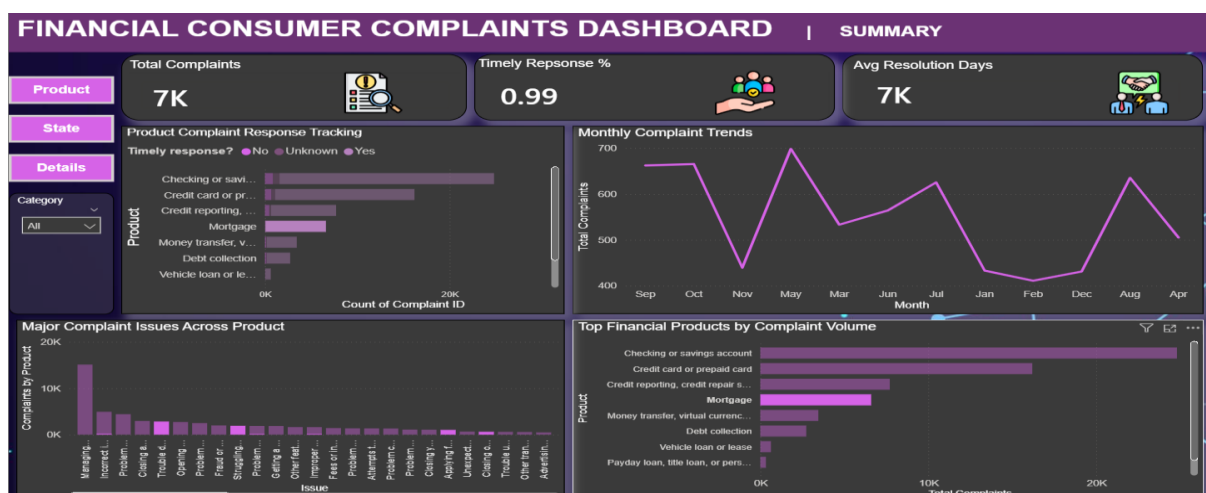
Navigation Buttons

Navigation buttons were implemented to create a multi-page dashboard architecture and improve user experience through seamless page navigation and analytical workflow management.



Cross-Filtering & Interactive Analysis

Cross-filtering functionality was enabled to support interactive exploratory analysis by dynamically filtering related visuals based on user selections and analytical interactions.



9. Insights & Conclusions

Summary Analysis Dashboard

Key Findings: The dashboard analysis identified high complaint concentration across specific financial products, recurring operational pain points, and region-wise variations in customer response performance through KPI-driven analytical monitoring.

Descriptive Analysis: Performed descriptive analytics to evaluate historical complaint distribution patterns across products, issues, states, and operational response metrics using interactive Power BI visualizations.

Diagnostic Analysis: Conducted diagnostic analysis to identify root-cause factors contributing to elevated complaint volumes, delayed response performance, and recurring customer issue categories.

Predictive Analysis: Leveraged trend and time-series analysis to identify recurring complaint behavior patterns and potential future operational risk areas across financial service segments.

Prescriptive Analysis: Recommended strategic process optimization, SLA-focused response improvement, and targeted operational interventions for high complaint-generating products and regions.

Key Insights

- **Specific** – Analysed financial consumer complaints across products, states, channels, and issue categories to identify key customer pain points and operational risks.
- **Measurable** – Measured business performance using KPI metrics such as total complaints, timely response rate, resolution efficiency, and complaint trends.
- **Achievable** – Built an interactive Power BI dashboard using Power Query, DAX, drillthrough pages, and dynamic visualizations for actionable analysis.
- **Relevant** – Supported customer experience improvement, compliance monitoring, and strategic business decision-making through insight-driven reporting.
- **Time-Bound** – Tracked complaint patterns and resolution performance over monthly and yearly periods to monitor business performance trends.

11. Product Analysis Dashboard

Key Findings: Total complaints recorded reached 63K, with Checking/Savings Accounts (~25K) contributing the highest complaint volume.

Credit Card & Prepaid Card services (~16K) emerged as the second-largest complaint category. The organization maintained a strong 94% timely response rate, but the average resolution period remained high at 61 days.

Descriptive Analysis: The dashboard highlights complaint distribution across financial products and monthly complaint trends.

Checking accounts consistently generated the highest complaint activity throughout the year. Complaint volume was heavily concentrated in banking and credit-related services.

Diagnostic Analysis: High complaint volumes in checking accounts indicate recurring operational and customer service issues.

Common complaint categories such as account management and payment-related issues

contributed significantly to dissatisfaction.

Longer resolution durations suggest inefficiencies in complaint closure workflows despite strong response performance.

Predictive Analysis: If current complaint trends continue, checking and credit card products are expected to remain the primary complaint drivers.

Complaint volume may increase during peak financial activity periods without proactive service improvements.

Delayed resolution cycles could negatively impact customer trust and satisfaction levels over time.

Prescriptive Analysis: Implement automated complaint escalation systems to reduce the current 61-day average resolution period.

Strengthen customer support processes for checking and credit-related services to reduce complaint recurrence.

Introduce proactive customer communication and self-service support features for faster issue resolution.

Key Insights:

- Approximately 65% of total complaints originated from checking accounts and credit card products, making them the most critical focus areas.
- The organization successfully maintained a 94% timely response rate, demonstrating strong complaint acknowledgment efficiency.
- Reducing complaint resolution time from 61 days to below 45 days can significantly improve customer experience and operational efficiency.

12. State Analysis Dashboard

Key Findings: California recorded the highest complaint volume with approximately 14K complaints, followed by Florida and Texas.

Most regional complaints were associated with account management and incorrect information issues.

Timely response performance remained stable at 94% across major states.

Descriptive Analysis: The dashboard visualizes complaint distribution across different states and regional trends over time.

California consistently showed higher complaint activity compared to other states throughout the reporting period.

Regional complaint concentration was primarily observed in densely populated financial markets.

Diagnostic Analysis: High complaint concentration in California and Florida indicates operational pressure in major service regions.

Account-related issues and payment problems emerged as major contributors to regional customer dissatisfaction.

Variations in complaint patterns suggest differences in regional service efficiency and customer handling processes.

Predictive Analysis: High-population states are likely to continue generating elevated complaint volumes if operational issues remain unresolved.

Complaint activity may increase in regions with high financial transaction density and customer dependency.

States with recurring complaint patterns may experience declining customer trust over time.

Prescriptive Analysis: Improve regional customer support accessibility in high-complaint states such as California and Florida.

Deploy state-wise complaint monitoring systems to identify and resolve recurring service issues faster.

Conduct regional operational audits to improve complaint handling consistency across locations.

Key Insights

- California alone contributed approximately 14K complaints, making it the highest complaint-generating region.
- More than 60% of total complaints originated from a limited number of major states.
- Regional service optimization initiatives should target measurable complaint reduction within the next 6 months.

13. Complaint-Level Detail Analysis

Key Findings: Detailed complaint records highlighted recurring issues related to closing checking/savings accounts.

Most complaint entries maintained a 100% timely response rate at the record level.

Complaint cases were distributed across multiple states including FL, TX, AZ, CA, and NY.

Descriptive Analysis: The dashboard provides complaint-level visibility using Complaint ID, Product, Issue, State, and Response metrics.

Checking and savings account complaints represented the most frequently occurring detailed cases.

Complaint records revealed repeated issue categories associated with account closure processes.

Diagnostic Analysis: Recurring account closure complaints indicate process inefficiencies and customer dissatisfaction in account management services.

Multiple complaints across different states suggest widespread operational challenges rather than isolated incidents.

Repeated issue patterns highlight the need for improved workflow standardization and customer guidance.

Predictive Analysis: Similar complaint categories may continue increasing if account closure workflows remain complex or unclear.

High-frequency complaint patterns may lead to greater customer churn and service dissatisfaction.

Operational bottlenecks in account handling processes may increase future complaint recurrence rates.

Prescriptive Analysis: Simplify account closure procedures and improve customer communication during account-related processes.

Introduce proactive issue-resolution mechanisms to prevent recurring customer complaints.

Strengthen complaint tracking systems for faster identification of repeated issue categories.

Key Insights :

- A significant share of detailed complaints were associated with checking/savings account closure issues.
 - Complaint records demonstrated 100% timely response performance for the analyzed complaint cases.
 - Streamlining account closure operations can help reduce recurring complaints and improve customer satisfaction within future reporting cycles.
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14. Conclusions

This project successfully transformed raw Financial Consumer Complaints data into an interactive Business Intelligence solution. The dashboard provides a scalable analytical framework for evaluating complaint behavior across products, regions, and operational response metrics. By integrating interactive filtering, data modeling, DAX-based KPI engineering, and multi-page analytical storytelling, the solution supports both executive-level reporting and operational-level diagnostic analysis.

Overall, the project demonstrates how Business Intelligence and data visualization techniques can be leveraged to improve complaint monitoring, enhance customer service evaluation, and support data-driven strategic decision-making.